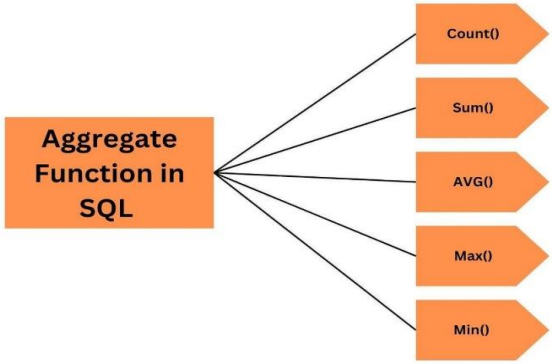
NYKAA BEAUTY STORE MANAGEMENT SYSTEM

SALES AND CUSTOMER ANALYTICS SYSTEM

1.Aggregate functions

Aggregate functions in SQL are used to perform calculations on multiple rows and return a single value as the result.



1.1COUNT()

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM Orders;

TOTAL_ORDERS
------------
           6
```

1.2SUM()

```
SQL> SELECT SUM(Total_Amount) AS TOTAL_SALES
  2  FROM Orders;

TOTAL_SALES
-----------
       6450
```

1.3AVG()

```
SQL> SELECT ROUND(AVG(Total_Amount),2) AS AVERAGE_SALES
  2  FROM Orders;

AVERAGE_SALES
-------------
         1075
```

1.4MIN()

```
SQL> SELECT MIN(Total_Amount) AS MINIMUM_SALES
  2  FROM Orders;

MINIMUM_SALES
-------------
          800
```

1.5MAX()

```
SQL> SELECT MAX(Total_Amount) AS MAXIMUM_SALES
  2  FROM Orders;

MAXIMUM_SALES
-------------
         1600
```

2.Generate Total Sales Reports

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_ORDERS,
  3      SUM(Total_Amount) AS TOTAL_SALES,
  4      ROUND(AVG(Total_Amount),2) AS AVERAGE_SALES,
  5      MIN(Total_Amount) AS MINIMUM_SALES,
  6      MAX(Total_Amount) AS MAXIMUM_SALES
  7  FROM Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES MINIMUM_SALES MAXIMUM_SALES
------------ ----------- ------------- ------------- -------------
           6        6450          1075           800          1600
```

2.1Customer-wise Sales Report

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      COUNT(O.Order_ID) AS TOTAL_ORDERS,
  5      SUM(O.Total_Amount) AS TOTAL_PURCHASE
  6  FROM Customer C
  7  JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  GROUP BY C.Customer_ID, C.Customer_Name
 10  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID CUSTOMER_NAME                 TOTAL_ORDERS TOTAL_PURCHASE
----------- ------------------------------ ------------ --------------
        101 Akshaya                                 2           1700
        104 Kavya                                   1           1600
        102 Priya                                   1           1200
        105 Swetha                                  1           1000
        103 Divya                                   1            950
```

3.Find Top customers based on purchase amount

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      SUM(O.Total_Amount) AS PURCHASE_AMOUNT
  5  FROM Customer C
  6  JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.Customer_Name
  9  ORDER BY PURCHASE_AMOUNT DESC
 10  FETCH FIRST 3 ROWS ONLY;

CUSTOMER_ID CUSTOMER_NAME                 PURCHASE_AMOUNT
----------- ------------------------------ ---------------
        101 Akshaya                                 1700
        104 Kavya                                   1600
        102 Priya                                   1200
```

4.identify best-selling products

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      SUM(OD.Quantity) AS TOTAL_QUANTITY
  5  FROM Product P
  6  JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY TOTAL_QUANTITY DESC;

PRODUCT_ID PRODUCT_NAME                               TOTAL_QUANTITY
---------- ---------------------------------------- --------------
       201 Face Wash                                               3
       205 Perfume                                                 3
       202 Moisturizer                                             2
       206 Foundation                                              2
       203 Lipstick                                                2
       204 Shampoo                                                 2
```

4.1product sales amount

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      SUM(OD.Quantity * OD.Unit_Price) AS SALES_AMOUNT
  5  FROM Product P
  6  JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY SALES_AMOUNT DESC;

PRODUCT_ID PRODUCT_NAME                               SALES_AMOUNT
---------- ---------------------------------------- ------------
       205 Perfume                                            2400
       206 Foundation                                         1200
       202 Moisturizer                                         800
       201 Face Wash                                           750
       203 Lipstick                                            700
       204 Shampoo                                             600

6 rows selected.
```

5.category-wise sales analysis

```
SQL> SELECT
  2      C.Category_ID,
  3      C.Category_Name,
  4      SUM(OD.Quantity * OD.Unit_Price) AS TOTAL_SALES
  5  FROM Category C
  6  JOIN Product P
  7  ON C.Category_ID = P.Category_ID
  8  JOIN Order_Details OD
  9  ON P.Product_ID = OD.Product_ID
 10  GROUP BY C.Category_ID, C.Category_Name
 11  ORDER BY TOTAL_SALES DESC;

CATEGORY_ID CATEGORY_NAME                    TOTAL_SALES
----------- ------------------------------ -----------
          4 Fragrance                             2400
          2 Makeup                                1900
          1 Skincare                              1550
          3 Haircare                               600
```

6.combined business sales report

```
SQL> SELECT
  2      (SELECT COUNT(*)
  3       FROM Customer) AS TOTAL_CUSTOMERS,
  4  (SELECT COUNT(*)
  5       FROM Orders) AS TOTAL_ORDERS,
  6   (SELECT COUNT(*)
  7       FROM Product) AS TOTAL_PRODUCTS,
  8   (SELECT SUM(Quantity)
  9       FROM Order_Details) AS PRODUCTS_SOLD,
 10   (SELECT SUM(Total_Amount)
 11       FROM Orders) AS TOTAL_SALES
 12  FROM DUAL;

TOTAL_CUSTOMERS TOTAL_ORDERS TOTAL_PRODUCTS PRODUCTS_SOLD TOTAL_SALES
--------------- ------------ -------------- ------------- -----------
              5            6              6            14        6450
```